

MANUEL JOSE MARTE

April 2026

Phipps 446
Johns Hopkins University School of Medicine
600 Wolfe Street
Baltimore, MD 21287, USA

 [ORCID](#)
 [Personal Website](#)
 [Google Scholar](#)
 [Bluesky](#)
 [Email](#)

EDUCATION

2025 **Ph.D., Speech, Language & Hearing Sciences**

Boston University

Dissertation: *A Multimodal and Naturalistic Approach to Assessment in Aphasia*

2016 **M.S., Communication Disorders**

State University of New York at New Paltz

2014 **B.A., Speech & Hearing Science**

University at Buffalo

RESEARCH EXPERIENCE

Department of Neurology, Johns Hopkins University School of Medicine

2025– **Postdoctoral Research Fellow**

[SCORE \(Stroke Cognitive Outcomes & REcovery\) Lab](#), PI: Argye E. Hillis

T32 NRSA Fellow, Kennedy Krieger Institute

- Built machine learning pipeline combining cortical parcellation volumes, white matter tract integrity, and lesion-network disconnection scores with acute clinical variables to predict 12-month language recovery trajectories
- Administering HD-tDCS to acute stroke patients in the Emergency Department as part of the TESSERACT-II precision tDCS clinical trial; managing REDCap protocol and session workflows
- Designing a waitlist randomized controlled trial using cerebellar neuromodulation guided by algorithmic prediction for treatment staging

Institute for Technology in Psychiatry, McLean Hospital, Harvard Medical School

2024–2025 **Senior Clinical Research Assistant**

[The Functional Neuroimaging & Bioinformatics Lab](#), PI: Einat Liebenthal

- Designed and implemented a multimodal naturalistic paradigm combining continuous affect ratings, eye-tracking, and movie-viewing to assess language-emotion coupling in post-stroke aphasia and mild cognitive impairment
- Recruited and tested 100+ participants (persons with aphasia, mild cognitive impairment, and neurotypical controls) across behavioral and neuroimaging sessions
- Developed automated multimodal media analysis integrating automatic speech recognition, LLM-based scoring, and acoustic-prosodic feature extraction for scalable narrative discourse assessment

Sargent College of Health & Rehabilitation Sciences, Boston University

2020–2025 **Doctoral Student Research Assistant**

Center for Brain Recovery, PI: Swathi Kiran

- Applied machine learning to predict naming recovery in bilingual post-stroke aphasia using dual-language proficiency, cognitive, and demographic features
- Quantified cross-language treatment generalization in bilingual and trilingual aphasia across multiple single-case and group designs
- Created and validated the LEX-BADAT open dataset characterizing language experience profiles in bilinguals with and without aphasia
- Analyzed social determinants of health associated with poststroke communication difficulties in U.S. Hispanic populations using NHIS data

State University of New York at New Paltz

2015–2016 **Graduate Student Research Assistant**

Department of Communication Disorders, PI: Dana Arthur & Inge Anema

- Investigated training effects on suprasegmental features in oral reading of non-native English speakers

University at Buffalo

2013–2014 **Undergraduate Student Research Assistant**

Center for Orofacial Pain Research, PI: Richard Ohrbach

- Examined effects of repeated anodal transcranial direct current stimulation of motor cortex on visuo-motor task performance

PUBLICATIONS

[‡] shared first authorship. ^{*} student mentee.

Peer-Reviewed Publications

1. Kiran, S., Carpenter, E., Grasemann, U., Scimeca, M., **Marte, M. J.**, Russell-Meill, M., Peñaloza, C., Tripodis, Y., & Miikkulainen, R. (2026). [Predicting bilingual aphasia treatment outcomes using digital twins: A double-blind randomized controlled trial.](#) *npj Digital Medicine*. doi: 10.1038/s41746-026-02583-9
2. Chen, X.J., **Marte, M. J.**, Kiran, S., & Blanco-Elorrieta, E. (2026). [Evidence for an integrated bilingual language system from discourse tasks in aphasia.](#) *Aphasiology*. doi: 10.1080/02687038.2026.2624735
3. Russell-Meill, M.[‡], **Marte, M. J.**[‡], Carpenter, E.[‡], & Kiran, S. (2025). [Navigating the Complexity of Bilingual Aphasia: Current Insights and Future Directions.](#) *Brain Sciences*. doi: 10.3390/brainsci15090989
4. Scimeca, M., Peñaloza, C., Carpenter, E., **Marte, M. J.**, Russell-Meill, M., & Kiran, S. (2025). [The evolution of word retrieval errors during semantic feature-based therapy in bilingual aphasia.](#) *Bilingualism: Language and Cognition*. doi: 10.1017/S1366728925100370
5. Russell-Meill, M., Carpenter, E., **Marte, M. J.**, Scimeca, M., Peñaloza, C., & Kiran, S. (2025). [Measurement of cross-language and cross-domain-generalization following semantic feature-based anomia therapy in bilingual aphasia.](#) *Neuropsychological Rehabilitation*. doi: 10.1080/09602011.2025.2522196
6. Peñaloza, C., **Marte, M. J.**, Billot, A., & Kiran, S. (2025). [Cross-language interactions during sequential anomia treatment in three languages: Evidence from a trilingual person with aphasia.](#) *Cortex*. doi: 10.1016/j.cortex.2025.05.017

7. **Marte, M. J.**, Carpenter, E., Scimeca, M., Russell-Meill, M., Peñaloza, C., Grasemann, U., Miikkulainen, R., & Kiran, S. (2025). [Machine learning predictions of recovery in bilingual post-stroke aphasia: Aligning insights with clinical evidence](#). *Stroke*. doi: 10.1161/STROKEAHA.124.047867
8. **Marte, M. J.**[‡], Russell-Meill, M.[‡], Carvalho, N.[‡], & Kiran, S. (2025). [Charting the Course of Aphasia Recovery: Factors, Trajectories, and Outcomes](#). *Annual Review of Linguistics*. doi: 10.1146/annurev-linguistics-011724-121245
9. Cordella, C., **Marte, M. J.**, Liu, H., & Kiran, S. (2025). [An introduction to machine learning for speech-language pathologists: Concepts, terminology, and emerging applications](#). *Perspectives of the ASHA Special Interest Groups: Sig 2 Neurogenic Communication Disorders*. doi: 10.1044/2024_PERSP-24-00037
10. **Marte, M. J.**, Addesso, D.^{*}, & Kiran, S. (2024). [Association Between Social Determinants of Health and Communication Difficulties in Poststroke U.S. Hispanic and Non-Hispanic White Populations](#). *American Journal of Speech-Language Pathology*. doi: 10.1044/2023_AJSLP-23-00232
11. **Marte, M. J.**, Peñaloza, C., & Kiran, S. (2023). [The cognate effect on lexical access in bilingual aphasia: Evidence from the Boston Naming Test](#). *Bilingualism: Language and Cognition*. doi: 10.1017/S1366728923000251
12. **Marte M. J.**[‡], Carpenter, E.[‡], Falconer, I., Scimeca, M., Abdollahi, F., Peñaloza, C. & Kiran, S. (2022). [LEX-BADAT: Language EXperience in Bilinguals With and Without Aphasia DATaset](#). *Frontiers in Psychology*. 13:875928. doi: 10.3389/fpsyg.2022.875928

Under Review

1. **Marte, M. J.**, Gillis, B., Wei, X., Galvin, C., Rigolo, L., Tie, Y., Kiran, S., Liebenthal, E. (*under review, Nature Communications*). [Real-time emotional valence judgements during movie-watching reveal impaired language-emotion coupling in aphasia](#).

In Preparation

1. **Marte, M. J.**, Chaves, M.^{*}, Kelly, L., Diaz-Carr, I., Neal, V., Stockbridge, M. D., & Hillis, A. E. (*in preparation*). Predicting 12-month aphasia resolution and connected-speech recovery from acute stroke data: Shared and distinct neural predictors.
2. **Marte, M. J.**, Lee, S., Wang, S., Goldin, K.^{*}, Girard, J., Varkanitsa, M., Tie, Y., Kiran, S., Liebenthal, E. (*in preparation*). Automated multimodal main concept analysis for narrative discourse in aphasia.
3. **Marte, M. J.**, Gillis, B., Tie, Y., Kiran, S., Liebenthal, E. (*in preparation*). Divergent visual attention strategies in aphasia and mild cognitive impairment: Evidence from naturalistic movie-viewing.
4. Carpenter, E., Scimeca, M., **Marte, M. J.**, Russell-Meill, M., Peñaloza, C., & Kiran, S. (*in preparation*). Moving away from parallel and differential language impairments in bilingual aphasia: A machine learning approach to the identification of bilingual aphasia clinical profiles.

GRANT SUPPORT

2026– NIH/NICHD [5T32HD007414-32](#)

Ruth L. Kirschstein Institutional National Research Service Award
Research Training in Rehabilitation for Brain Injury and Neurological Disability
 PI(s): Amy J. Bastian, Christopher Thompson
Role: Postdoctoral Trainee (NRSA Fellow)

2024–2025 NIH/NIDCD [3R01DC020965-02S1](#)

Research Supplements to Promote Diversity in Health-Related Research

Naturalistic Neuroimaging for Presurgical Language Mapping

PI(s): Yanmei Tie, Einat Liebenthal

Role: Diversity Supplement Recipient

2023–2024 NIH/NIDCD [5R01DC020965](#)

Naturalistic Neuroimaging for Presurgical Language Mapping

PI(s): Yanmei Tie, Einat Liebenthal

Role: Graduate Research Assistant

2020–2022 NIH/NIDCD [5T32DC013017-05](#)

Ruth L. Kirschstein Institutional National Research Service Award

Advanced Research Training in Communication Sciences and Disorders

PI(s): Christopher A. Moore

Role: Predoctoral Trainee (NRSA Fellow)

HONORS & AWARDS

2026 *ASHA Pathways Program*

[ASHA Academic & Research Mentoring \(ARM\) Network](#)

2025– *Neuroscience Scholars Program Associate*

Society for Neuroscience

2025 *NIH/NIDCD Young Investigator Travel Fellowship Award*

2025 Academy of Aphasia

Mentor: Brielle Stark

NIH/NIDCD RSCA Student Fellowship Award

2025 Clinical Aphasiology Conference

Mentor: Brielle Stark

2024 *ASHA Research Mentoring-Pair Travel Award*

American Speech-Language-Hearing Association Convention 2024

Mentor: Stacy Harnish

2023 *NIH/NIDCD Young Investigator Travel Fellowship Award*

2023 Academy of Aphasia

Mentor: Marco Calabria

NIH/NIDCD RSCA Student Fellowship Award

2023 Clinical Aphasiology Conference

Mentor: Heather Dial

2022 *NIH/NIDCD Student Scholar Award*

16th Annual Eleanor M. Saffran Conference on Cognitive Neuroscience and Rehabilitation of Communication Disorders

2021 *NSF Research Traineeship Program–Understanding the Brain: Neuropotonics*

[DGE-1633516](#)

2014 *Center for Undergraduate Research & Creative Activities Grant*

University at Buffalo

INVITED TALKS

- 2025 **Marte, M. J.** Context matters: From individual differences to naturalistic communication in post-stroke aphasia. *SLHS Colloquium*, University of Arizona, Tucson, AZ, USA.
- 2024 **Marte, M. J.** Two languages, one brain: Understanding bilingual aphasia; Dos idiomas, un cerebro: Comprensión de la afasia bilingüe. *Charlas Científicas*, University of Arizona, Tucson, AZ, USA.
- 2023 **Marte, M. J.** Predicting treatment response in bilingual aphasia: An exploratory machine learning approach. *T32 Site Visit*, Boston University, Boston, MA, USA.
- 2022 **Marte, M. J.**, Carpenter, E., Scimeca, M., Kiran, S. Insights into bilingual aphasia from the PROCoM clinical trial. *T32 Clinical Connections Seminar*, University of Connecticut, Storrs, CT, USA.

PEER-REVIEWED POSTERS & PRESENTATIONS

‡ shared first authorship. * student mentee.

Oral Presentations

1. Stockbridge, M. D., Meier, E. L., & **Marte, M. J.** (2026, May). Aphasia taxonomy and linguistic exceptionalism in the Information Age: Shifting cultural perspectives. *Clinical Aphasiology Conference 2026*, Pittsburgh, PA, USA. *Roundtable*.
2. **Marte, M. J.**, Gillis, B., Galvin, C., Rigolo, L., Tie, Y., Kiran, S., & Liebenthal, E. (2024, December). Emotional processing in post-stroke aphasia: Insights from moment-by-moment valence ratings during movie-watching. *American Speech-Language-Hearing Association Convention 2024*, Seattle, WA, USA.
3. Scimeca, M.‡, Carpenter, E.‡, **Marte, M. J.**‡, & Kiran, S. (2024, March) Predicting language recovery for bilingual individuals with aphasia. *Brain Injury Association of Massachusetts 2024 Conference*, Marlborough, MA, USA.
4. **Marte, M. J.**, Addesso, D.* & Kiran, S. (2023, May). Social determinants of health and poststroke communication difficulties in Hispanic and non-Hispanic white adults. *Clinical Aphasiology Conference 2023*, Atlantic City, NJ, USA.
5. **Marte, M. J.**, Scimeca, M., Addesso, D.* & Kiran, S. (2022, November). Hispanic Stroke Care in the United States: Insights from Epidemiology, Health Disparity, and Clinical Practice. *American Speech-Language-Hearing Association Convention 2022*, New Orleans, LA, USA.

Poster Presentations

1. **Marte, M. J.**, Chaves, M.*, Kelly, L., Diaz-Carr, I., Neal, V., Faria, A. V., Stockbridge, M. D., & Hillis, A. E. (*submitted*). Predicting Language Recovery at 12 Months from Acute Stroke Data: Machine Learning Identifies Shared and Distinct Neural Predictors. *American Neurological Association 2026 Annual Meeting*, San Diego, CA, USA.
2. Zain, N., **Marte, M. J.**, Ray, S., Stevens, R. D., & Stockbridge, M. D. (*submitted*). Mapping Ischemic Stroke with Speech-Language Features: A Machine Learning Analysis of Multimodal Picture Description Data. *American Neurological Association 2026 Annual Meeting*, San Diego, CA, USA.
3. **Marte, M. J.**, Lee, S., Wang, S., Goldin, K.*, Girard, J., Varkanitsa, M., Tie, Y., Kiran, S., Liebenthal, E. (2026, March). Assessing spoken discourse in aphasia using multimodal artificial intelligence. *39th Annual Conference on Human Sentence Processing*, Cambridge, MA, USA.

4. Wei, X., **Marte, M. J.**, Rigolo, L., Galvin, C., Golby, A., Tie, Y., & Liebenthal, E. (2025, November). Individual differences of behavior show in dynamic brain response in movie watching. *Society for Neuroscience Conference 2025*, San Diego, CA, USA.
5. **Marte, M. J.**, Gillis, B., Goldin, K.^{*}, Wei, X., Galvin, C., Rigolo, L., Tie, Y., Kiran, S., & Liebenthal, E. (2025, October). Multimodal analysis of linguistic, emotional, and visual attention processing of naturalistic movie stimuli in persons with aphasia. *Academy of Aphasia 63rd Annual Meeting*, San Diego, CA, USA.
6. **Marte, M. J.**, Gillis, B., Goldin, K.^{*}, Wei, X., Galvin, C., Rigolo, L., Tie, Y., Kiran, S., & Liebenthal, E. (2025, September). Multimodal signatures in aphasia: A convergent analysis of emotion, visual attention, and language during movie-viewing. *Science of Aphasia 2025*, Copenhagen, Denmark.
7. Goldin, K.^{*}, **Marte, M. J.**, Gillis, B., Wei, X., Galvin, C., Rigolo, L., Tie, Y., Kiran, S., & Liebenthal, E. (2025, July). Errors in Automatic Speech Recognition Transcripts in Neurotypical Adults and Individuals with Post-Stroke Aphasia From a Naturalistic Experiment. *Neuroscience of the Everyday World Conference 2025*, Boston, MA, USA.
8. **Marte, M. J.**, Gillis, B., Goldin, K.^{*}, Wei, X., Galvin, C., Rigolo, L., Tie, Y., Kiran, S., & Liebenthal, E. (2025, July). Continuous Valence Ratings During Naturalistic Movie-Viewing Reveal Heterogeneous Language-Emotion Integration in Aphasia. *Neuroscience of the Everyday World Conference 2025*, Boston, MA, USA.
9. Wei, X., **Marte, M. J.**, Gillis, B., Galvin, C., Rigolo, L., Golby, A., Tie, Y., & Liebenthal, E. (2025, June). Understanding Brain Dynamics During Movie Watching. *Organization for Human Brain Mapping 2025 Annual Meeting*, Brisbane, Australia.
10. **Marte, M. J.**, Gillis, B., Faris, R.^{*}, Galvin, C., Rigolo, L., Tie, Y., Kiran, S., & Liebenthal, E. (2025, June). The clinical utility of naturalistic movie-viewing: Combining emotional reactivity, gaze patterns, and contextual language processing. *Organization for Human Brain Mapping 2025 Annual Meeting*, Brisbane, Australia.
11. **Marte, M. J.**, Gillis, B., Faris, R.^{*}, Galvin, C., Rigolo, L., Tie, Y., Kiran, S., & Liebenthal, E. (2025, May). Movie-viewing reveals novel patterns of emotional and visual processing in post-stroke aphasia. *Clinical Aphasiology Conference 2025*, Albuquerque, NM, USA.
12. **Marte, M. J.**, Gillis, B., Faris, R.^{*}, Galvin, C., Rigolo, L., Tie, Y., Kiran, S., & Liebenthal, E. (2025, March). Naturalistic movie-viewing paradigms reveal distinct patterns of cognitive-linguistic processing across clinical populations. *Cognitive Neuroscience Society 2025 Annual Meeting*, Boston, MA, USA.
13. Russell-Meill, M., Carpenter, E., **Marte, M. J.**, Scimeca, M., Peñaloza, C., & Kiran, S. (2024, December). Outcomes of Semantic Feature-Based Therapy in Bilingual Aphasia: Patterns of Generalization Across Languages. *American Speech-Language-Hearing Association Convention 2024*, Seattle, WA, USA.
14. Russell-Meill, M., Carpenter, E., **Marte, M. J.**, Scimeca, M., Peñaloza, C., & Kiran, S. (2024, October). Beyond Naming: Promoting Generalization in Bilingual Aphasia with Semantic Feature-Based Treatment. *Academy of Aphasia 62nd Annual Meeting*, Nara, Japan.
15. Galvin, C., Rigolo, L., **Marte, M. J.**, Gillis, B., Tie, Y., Kiran, S., Liebenthal, E., & Golby, A. (2024, October). Developing a movie clip database for naturalistic brain mapping. *Society for Neuroscience Conference 2024*, Chicago, IL, USA.
16. **Marte, M. J.**, Gillis, B., Galvin, C., Rigolo, L., Tie, Y., Kiran, S., & Liebenthal, E. (2024, October). Dynamics of emotion processing in post-stroke aphasia: Insights from continuous valence ratings during naturalistic movie viewing. *Society for Neuroscience Conference 2024*, Chicago, IL, USA.
17. Gillis, B., **Marte, M. J.**, Galvin, C., Rigolo, L., Tie, Y., Kiran, S., & Liebenthal, E. (2024, August). Language and emotion processing dynamics in post-stroke aphasia: Continuous and holistic valence ratings of naturalistic movie-viewing. *Neuroscience of the Everyday World 2024*, Boston, MA, USA.

18. Scimeca, M., Carpenter, E., **Marte, M. J.**, Russell, M., Peñaloza, C., & Kiran, S. (2024, May) Assessing assessment in bilingual aphasia: Towards cultural-linguistic equity in clinical testing. *Clinical Aphasiology Conference 2024*, Big Island of Hawai'i, HI, USA.
 19. Carpenter, E., Grasemann, U., Scimeca, M., **Marte, M. J.**, Russell, M., Peñaloza, C., Miikkulainen, R., & Kiran, S. (2024, May) A randomized controlled trial examining the predictive power of computational simulations on treatment-induced naming outcomes in Spanish-English bilinguals with aphasia. *Clinical Aphasiology Conference 2024*, Big Island of Hawai'i, HI, USA.
 20. Carpenter, E., Grasemann, U., Scimeca, M., **Marte, M. J.**, Russell, M., Peñaloza, C., Miikkulainen, R., & Kiran, S. (2024, May). Simulating therapy-induced recovery in bilingual aphasia via computational modeling and machine learning approaches. *13th World Congress for Neurorehabilitation*, Vancouver, Canada.
 21. Carpenter, E., Braun, E. J., **Marte, M. J.**, Yücel, M. A., Boas, D. A., & Kiran, S. (2023, October). fNIRS Functional Connectivity in Chronic Post-Stroke Aphasia: A Pilot Study. *15th Annual Meeting of the Society for the Neurobiology of Language*, Marseille, France.
 22. **Marte, M. J.**, Carpenter, E., Scimeca, M., Russell, M., Falconer, I., Peñaloza, C., & Kiran, S. (2023, October). Predicting treatment responsiveness in bilingual aphasia: The role of dual language information. *Academy of Aphasia 61st Annual Meeting*, Reading, England.
 23. Carpenter, E., **Marte, M. J.**, Scimeca, M., Russell, M., Falconer, I., Peñaloza, C., & Kiran, S. (2023, October). Factors influencing language impairment in bilingual aphasia. *Academy of Aphasia 61st Annual Meeting*, Reading, England.
 24. Braun, E. J., Carpenter, E., Gao, Y., **Marte, M. J.**, Yücel, M. A., Boas, D. A., & Kiran, S. (2023, May). Cortical activity patterns for narrative production in individuals with and without aphasia via functional near-infrared spectroscopy (fNIRS). *Clinical Aphasiology Conference 2023*, Atlantic City, NJ, USA.
 25. Scimeca, M., Peñaloza, C., Carpenter, E., **Marte, M. J.**, Falconer, I., Combs, A., Gray, T., Shamapant, S., & Kiran, S. (2022, December). Telerehabilitation for language deficits in Spanish-English individuals with post-stroke aphasia. *12th World Congress for Neurorehabilitation*, Vienna, Austria.
 26. Carpenter, E., Braun, E. J., **Marte, M. J.**, Scimeca, M., Yücel, M. A., Boas, D. A., & Kiran, S. (2022, October). Neuroscience in the Everyday World: Cortical Activation during Computer-Based Conversation Tasks Using Integrated fNIRS-EEG. *14th Annual Meeting of the Society for the Neurobiology of Language*, Philadelphia, PA, USA.
 27. Addesso, D.* , **Marte, M. J.**, Abdollahi, F., & Kiran, S. (2022, September). Social determinants of health contributing to disparities in stroke-associated variables in Hispanic populations. *16th Annual Eleanor M. Saffran Conference on Cognitive Neuroscience and Rehabilitation of Communication Disorders*, Philadelphia, PA, USA.
- *Selected for NIH/NIDCD Student Scholar Award.
28. **Marte, M. J.**, Addesso, D.* , Abdollahi, F., & Kiran, S. (2022, September). U.S. Hispanic ethnicity and stroke- and neurorehabilitation-associated outcomes: Disparity-driven heterogeneity. *16th Annual Eleanor M. Saffran Conference on Cognitive Neuroscience and Rehabilitation of Communication Disorders*, Philadelphia, PA, USA.
 29. Scimeca, M., Carpenter, E., **Marte, M. J.**, Combs, A., & Kiran, S. (2022, September). Enabling access to post-stroke rehabilitation in Hispanic bilinguals with aphasia: Lessons learned and challenges ahead. *16th Annual Eleanor M. Saffran Cognitive Neuroscience Conference*, Philadelphia, PA, USA.
 30. **Marte, M. J.**, Billot, A., Peñaloza, C., & Kiran, S. (2022, June). Cross-linguistic effects of a semantic treatment for anomia provided in three languages: Evidence from multilingual aphasia. *International Aphasia Rehabilitation Conference 2022*, Philadelphia, PA, USA.

31. Braun, E. J., Gao, Y., Carpenter, E., **Marte, M. J.**, Scimeca, M., Abdollahi, F., Yücel, M. A., Boas, D. A., & Kiran, S. (2021, October). Use of fNIRS to investigate cortical activation for conversational responses and narrative discourse. *Society for functional Near-Infrared Spectroscopy 2021*, virtual conference.
32. **Marte, M. J.** & Kiran, S. (2021, October). Influences on cognate facilitation in healthy bilinguals and bilinguals with aphasia. *13th Annual Meeting of the Society for the Neurobiology of Language*, virtual conference.
33. **Marte, M. J.** & Anema, I. (2015, May). Training Effects on Suprasegmental Features in Oral Reading of Non-Native English Speakers. *10th International Symposium on Bilingualism*, New Brunswick, NJ, USA.
34. Oldani, P., Hogan, C., Sauvageau, A., **Marte, M. J.**, Rudz, D., Agulia, B., Hanley, S., Quinn, T., Sharma, S., & Scerra, N., Ohrbach, R. (2014, November). The effects of varying levels of repeated anodal transcranial direct current stimulation (tDCS) of the motor cortex on visuo-motor task performance. *Biomedical Engineering Society 2014 Annual Meeting*, San Antonio, TX, USA.

TEACHING & MENTORING

Workshops

- 2024 *Introduction to PsychoPy for Experimental Research*.
Center for Brain Recovery, Boston University
Designed and led hands-on workshop with demo scripts for building psycholinguistic experiments in PsychoPy
- 2023 *Machine Learning Approaches to Analyzing Brain and Health Data*.
Hariri Institute, Boston University
Developed and presented live demos for designing supervised ML pipelines applied to medical research data
- 2021 *Principal Component Analysis in Aphasia Research*.
Aphasia Research Laboratory, Boston University
Didactic and applied session on PCA for dimensionality reduction of language assessments and language use questionnaires in aphasia research

Guest Lectures

- 2025 *Clinical assessment and intervention in bilingual aphasia*.
SLHS 535 Bilingualism, Multiculturalism, and Language Varieties
University of Arizona
- 2024 *Navigating clinical and scientific training in speech-language pathology*.
SLHS 756 Right Hemisphere Syndrome, TBI, Dementia in Communicative Disabilities
San Francisco State University
- 2023 *Navigating clinical and scientific training in speech-language pathology*.
SLHS 756 Right Hemisphere Syndrome, TBI, Dementia in Communicative Disabilities
San Francisco State University

Assessment and intervention in bilingual adults with aphasia.
SAR SH725 Bilingual Considerations in Language Assessment and Intervention
Boston University

Topics in bilingual aphasia.
CD 690 Aphasia
Emerson College

2022 *Introduction to bilingual aphasia.*

SAR SH 548 Introduction to Language Disorders Across the Lifespan
Boston University

Topics in bilingual aphasia.

CD 690 Aphasia (× 2)

Emerson College

2021 *Topics in bilingual aphasia.*

CD 690 Aphasia

Emerson College

Co-instructor

2016 CMD 301 Speech Science

State University of New York at New Paltz

Co-taught lectures, led laboratory and discussion sections

Teaching Assistant

2015 CMD 301 Speech Science

CMD 305 Hearing Science

CMD 310 Anatomy & Physiology of the Speech and Hearing Mechanism

State University of New York at New Paltz

Led laboratory and discussion sections, facilitated review sessions, graded coursework

2014 CMD 305 Hearing Science

CMD 310 Anatomy & Physiology of the Speech and Hearing Mechanism

State University of New York at New Paltz

Led laboratory and discussion sections, facilitated review sessions, graded coursework

Tutor

2014–2015 CMD 301 Speech Science, CMD 305 Hearing Science, CMD 310 Anatomy & Physiology of the Speech and Hearing Mechanism

State University of New York at New Paltz

Research Mentorship

^{UG} undergraduate. ^G graduate. ^{HS} high school.

2026– Mathew Chaves (Johns Hopkins '26)^{UG} — Lesion-network prediction of discourse recovery; co-authored poster at American Neurological Association 2026

2025– Rachel Kollarmalil (Johns Hopkins '30)^{UG}

2025– Vandie Dumaboc (Harvard University '27)^{UG}

2025 Kari Goldin (University of Massachusetts Amherst '26)^{UG} — ASR error analysis in aphasia; co-authored posters at Academy of Aphasia 2025 and Neuroscience of the Everyday World 2025

2024–2025 Isabel Yu (Boston University '27)^{UG}

2024–2025 Rowan Faris (University of Massachusetts Amherst '27)^{UG} — Co-authored posters at OHBM 2025, CAC 2025, and CNS 2025

2023–2025 Conversation, Health, Art, Technology Group Supervisor

2022–2023 Sofia Garcia Martinez (University of Illinois at Chicago '23)^G

M.S. in Biomedical Visualization, Master's Thesis Deliverable [Sample]

- 2021–2023 David Addesso (Boston University '22)^{UG}
UROP Mentee — Co-authored publication in *AJSLP* and presentations at Saffran Conference 2022; selected for NIH/NIDCD Student Scholar Award
- 2021 Hongli Zheng (Boston University '22)^{UG}
UROP Mentee
- 2021 Emily Benson-Tyler (Port Washington High School '22)^{HS}
Research in Science & Engineering Program Mentee; Regeneron Science Talent Search

SERVICE

Service to the Field

Ad hoc reviewer (41+ reviews): *Aphasiology*, *American Journal of Speech-Language Pathology*, *Behavioral Sciences*, *BioMedical Engineering OnLine*, *Brain Communications*, *Clinical Linguistics & Phonetics*, *Cognitive Processing*, *Cortex*, *Digital Health*, *Discover Neuroscience*, *International Journal of Multilingualism*, *JAMA Neurology*, *Journal of Communication Disorders*, *Journal of Speech, Language, and Hearing Research*, *NeuroImage: Clinical*, *Neurological Research*, *Neurorehabilitation and Neural Repair*, *npj Artificial Intelligence*, *npj Dementia*, *Scientific Data*, *Scientific Reports*

- 2026 Reviewer, Provost's Undergraduate Research Award, Hopkins Office for Undergraduate Research
- 2022 Session Chair, BIG BRAIN WORKSHOP 2022

Service to the Community

- 2025 Judge, Youth 4 Neuro
- 2025 Volunteer, Neuroscience of the Everyday World 2025
- 2023 Volunteer, Neuroscience of the Everyday World 2023
- 2022 Judge, Massachusetts High School Science & Engineering Fair
- 2021 Talk and Poster Moderator, 46th Annual Boston University Conference on Language Development

Professional Memberships

Society for Neuroscience
 Cognitive Neuroscience Society
 ASHA SIG 2: Neurogenic Communication Disorders
 ASHA SIG 14: Cultural and Linguistic Diversity

PROFESSIONAL DEVELOPMENT

- 2025 New England Future Faculty Workshop
- 2025 Oxford Clinical Neuroimaging Course, Wellcome Centre for Integrative Neuroimaging
- 2024 NIH Diversity Supplement Professional Development and Networking Workshop
- 2023 ASHA Mentoring Academic-Research Careers (MARC)
- 2022 ASHA Minority Student Leadership Program (MSLP)
- 2022 ASHA Student to Empowered Professional (S.T.E.P.)
- 2021 National Science Foundation Research Traineeship Program
 Understanding the Brain: Neurophotonics

2018 Foundations in Myofascial Release for Neck, Voice, and Swallowing Disorders
Pain Relief Center

SELECTED CLINICAL EXPERIENCE

2016–2020 *Northeast Center for Rehabilitation and Brain Injury*, NY, USA.

Speech–Language Pathologist and Practicum Supervisor. Evaluation and treatment services for adults with augmentative and alternative communication needs, ventilator dependency, executive dysfunction, pragmatic deficits, short- and long-term memory deficits, disorders of consciousness, voice, language, motor speech, auditory processing, neurogenic fluency, cognitive communication, and feeding and swallowing disorders. Mentoring and oversight of graduate-level student clinicians in this setting.

2018–2020 *Ghent Rehabilitation and Nursing Center*, NY, USA

Speech–Language Pathologist. Evaluation and treatment services for older adults with dementia, short- and long-term memory deficits, language, cognitive communication, and feeding and swallowing disorders.

PROFESSIONAL LICENSES AND CERTIFICATIONS

Certificate of Clinical Competence from the American Speech–Language–Hearing Association
Licensed Speech–Language Pathologist, State of Maryland

MEDIA

2026 Futurism at JHU Podcast: *Decoding Aphasia and Recovery From Stroke*.

2025 American Heart Association News: *Speaking more than one language can add layers to stroke recovery*.

SKILLS

Languages: English (native), Spanish (native).

Programming: Python, R, $\text{\LaTeX}$, shell scripting.